Surds - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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Question 1 *Practice question written for this worksheet (not from a past paper)*

Write $\sqrt{18}$ in the form $a\sqrt{2}$.

$\sqrt{9 \times 2} = 3\sqrt{2}$

Answer: 3√2

Question 2 *Practice question written for this worksheet (not from a past paper)*

Write $\sqrt{98}$ in the form $a\sqrt{2}$.

$\sqrt{49 \times 2}$

Answer: 7√2

Question 3 *Practice question written for this worksheet (not from a past paper)*

Simplify $\sqrt{72}$

$\sqrt{36 \times 2}$

Answer: 6√2

Question 4 *Practice question written for this worksheet (not from a past paper)*

Simplify $2\sqrt{3} + 5\sqrt{3}$

Add like surds

Answer: 7√3

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Simplify $\sqrt{50} + \sqrt{8}$

$5\sqrt{2} + 2\sqrt{2}$

Answer: 7√2

Question 6 *Practice question written for this worksheet (not from a past paper)*

Write $5\sqrt{6}$ as a single square root.

$5 = \sqrt{25}$; $\sqrt{25 \times 6}$

Answer: √150

Question 7 *Practice question written for this worksheet (not from a past paper)*

Rationalise the denominator of $\frac{10}{\sqrt{5}}$

Multiply top and bottom by $\sqrt{5}$: $\frac{10\sqrt{5}}{5}$

Answer: 2√5

Question 8 *Practice question written for this worksheet (not from a past paper)*

Rationalise the denominator of $\frac{5 + \sqrt{5}}{\sqrt{5}}$

$\frac{5\sqrt{5} + 5}{5}$

Answer: √5 + 1

Question 9 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(2 + \sqrt{5})^2$

$4 + 2\sqrt{5} + 2\sqrt{5} + 5$

Answer: 9 + 4√5

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Question 10 *Practice question written for this worksheet (not from a past paper)*

Write $\frac{6}{3+\sqrt{3}}$ in the form $a-\sqrt{b}$.

Multiply by $\frac{3-\sqrt{3}}{3-\sqrt{3}}$: $\frac{6(3-\sqrt{3})}{9-3}$

Answer: 3 - √3

Question 11 *Practice question written for this worksheet (not from a past paper)*

Simplify $\frac{\sqrt{125}+\sqrt{80}}{\sqrt{5}}$

$\sqrt{125}=5\sqrt{5}$, $\sqrt{80}=4\sqrt{5}$

$\frac{9\sqrt{5}}{\sqrt{5}}$

Answer: 9

Question 12 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2022 · Question 13*

13	$\sqrt{8}+4-(\sqrt{8}-4)(=8)$ **and** $\sqrt{8}+4+(\sqrt{8}-4)(=2\sqrt{8}=4\sqrt{2})$	$(a+b)(a-b)=a^2-b^2$ **and** $(\sqrt{8}+4)^2-(\sqrt{8}-4)^2$		3	M1 for correct substitutions into expression for $a+b$ **and** $a-b$ **or** expand the expression to get a^2-b^2 **and** substitute into this expression.
	('8')('2$\sqrt{8}$') or $\sqrt{2048}$ or $16\sqrt{8}$ or $32\sqrt{2}$ or $8\sqrt{32}$ or $8\sqrt{8\times4}$ oe				M1 (dep M1)
			8		A1 (dep both M marks)
					Total 3 marks

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Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 16

Q	Working	Answer	Mark	Notes
	$\frac{12}{\sqrt{2}-1}\times\frac{\sqrt{2}+1}{\sqrt{2}+1}$ or $\frac{12}{\sqrt{2}-1}\times\frac{-\sqrt{2}-1}{-\sqrt{2}-1}$ and $4\sqrt{2}$ or $2\sqrt{8}$ or $\sqrt{32}$ oe		3	M1 for showing a correct method for rationalising the denominator and dealing with $\left(\sqrt{2}\right)^5$
	E.g. $12\sqrt{2}+12-4\sqrt{2}$ or $8\sqrt{2}+12$ $12\sqrt{2}+12-2\sqrt{8}$ or $12\sqrt{2}+12-\sqrt{32}$ oe			M1 dep expression must be in surd form
	E.g. $12\sqrt{2}(+12)-4\sqrt{2}=8\sqrt{2}(+12)=$ $2\sqrt{4^2\times2}(+12)=2\sqrt{32}(+12)$ or $12\sqrt{2}(+12)-2\sqrt{8}=6\sqrt{8}(+12)-2\sqrt{8}=4\sqrt{8}(+12)=$ $2\sqrt{4\times8}(+12)=2\sqrt{32}(+12)$ or $12\sqrt{2}(+12)-\sqrt{32}=3\sqrt{4^2\times2}(+12)-\sqrt{32}=$ $2\sqrt{32}(+12)$ oe Note $8\sqrt{2}=2\sqrt{4^2\times2}$ or $2\sqrt{16\times2}$ or $\sqrt{32\times4}$ or $\sqrt{64\times2}$ $12\sqrt{2}=3\sqrt{4^2\times2}$ or $3\sqrt{16\times2}$ or $\sqrt{32\times9}$	Shown		A1 dep on M2 for showing working to given answer (they may dismiss the +12 and just deal with the surd part for this stage)
				Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 16

16		$\frac{4}{3\sqrt{5}+7}$, $\frac{3\sqrt{5}-7}{3\sqrt{5}-7}$ **or** $\frac{4}{3\sqrt{5}+7}$, $\frac{-3\sqrt{5}+7}{-3\sqrt{5}+7}$ oe		3	M1 for multiplying the numerator and denominator by $3\sqrt{5}-7$ or $-3\sqrt{5}+7$ (may be implied)
		eg $\frac{4(3\sqrt{5}-7)}{45-21\sqrt{5}+21\sqrt{5}-49}$ **or** $\frac{4(3\sqrt{5}-7)}{45-7^2}$ **or** $\frac{12\sqrt{5}-28}{45-21\sqrt{5}+21\sqrt{5}-49}$ **or** $\frac{12\sqrt{5}-28}{45-7^2}$ **or** $\frac{4(3\sqrt{5}-7)}{45-49}$ **or** $\frac{4(3\sqrt{5}-7)}{-4}$ **or** $\frac{12\sqrt{5}-28}{45-49}$ **or** $\frac{12\sqrt{5}-28}{-4}$			M1 for expanding the denominator in a correct fraction denominator may be 4 terms which all need to be correct $\frac{4}{3\sqrt{5}+7}$, $\frac{3\sqrt{5}-7}{3\sqrt{5}-7}=7-3\sqrt{5}$ scores M1M0 Implies the 1st mark
		Working required	$7-\sqrt{45}$		A1 dep on M2 SCB1 for answer $7-\sqrt{45}$ with no method marks awarded SCB2 for $7-\sqrt{45}$ if you would award the 1st M1 but not the 2nd M1 (total 2 marks)
					Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 16

16	$\frac{2\sqrt{3}}{\sqrt{3}-1}\times\frac{\sqrt{3}+1}{\sqrt{3}+1}$ or $\frac{2\sqrt{3}}{\sqrt{3}-1}\times\frac{-\sqrt{3}-1}{-\sqrt{3}-1}$		3	M1 for explicitly multiplying the numerator and the denominator by $\sqrt{3}+1$ or $-\sqrt{3}-1$
	$\frac{2\times3+2\sqrt{3}}{3-1}$ or $\frac{6+2\sqrt{3}}{3-1}$ or $\frac{6+2\sqrt{3}}{2}$ oe $\frac{-2\times3-2\sqrt{3}}{-3+1}$ or $\frac{-6-2\sqrt{3}}{-3+1}$ or $\frac{-6-2\sqrt{3}}{-2}$ oe			M1 dep on M1 (numerator expanded for 2 terms which need to be all correct and denominator may be 4 terms which need to be all correct)
	Working required	$3+\sqrt{3}$		A1 allow $\sqrt{3}+3$ (dep on M2)
				Total 3 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2019 · Question 16

16		$\frac{4+\sqrt{8}}{\sqrt{2}-1}\times\frac{(\sqrt{2}+1)}{(\sqrt{2}+1)}$			M1 for rationalising the denominator by multiplying numerator and denominator by $\sqrt{2}+1$ (or $-\sqrt{2}-1$) condone missing brackets
		e,g, $\frac{4\sqrt{2}+4+\sqrt{8}\sqrt{2}+\sqrt{8}}{2-1}$ **or** $\frac{4\sqrt{2}+4+4+\sqrt{8}}{2-1}$ **or** $\frac{4\sqrt{2}+4+\sqrt{16}+\sqrt{8}}{2-1}$ **or** $=4\sqrt{2}+4+4+\sqrt{8}$ oe			M1 (dep) for expansion of numerator with at least 3 terms correct oe Using $-\sqrt{2}-1$ e.g. $\frac{-4\sqrt{2}-4-\sqrt{8}\sqrt{2}-\sqrt{8}}{-2+1}$ **or** $\frac{-4\sqrt{2}-4-4-\sqrt{8}}{-2+1}$ **or** $\frac{-4\sqrt{2}-4-\sqrt{16}-\sqrt{8}}{-2+1}$
			$8+6\sqrt{2}$	3	A1 (dep on M2) or for stating $a = 8$ and $b = 6$
					Total 3 marks

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Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 17*

Q	Working	Answer	Mark	Notes
	$\frac{2\sqrt{7}+2}{\sqrt{7}-3}\times\frac{\sqrt{7}+3}{\sqrt{7}+3}$ **or** $\frac{2\sqrt{7}+2}{\sqrt{7}-3}\times\frac{-\sqrt{7}-3}{-\sqrt{7}-3}$		3	M1 for explicitly multiplying the numerator and denominator by $\sqrt{7}+3$ or $-\sqrt{7}-3$
	$\frac{2\sqrt{7}\sqrt{7}+6\sqrt{7}+2\sqrt{7}+6}{\sqrt{7}\sqrt{7}+3\sqrt{7}-3\sqrt{7}-9}$ oe **or** $\frac{14+6\sqrt{7}+2\sqrt{7}+6}{7-9}$ oe **or** $\frac{20+8\sqrt{7}}{-2}$ **or** $\frac{20+\sqrt{448}}{-2}$ or $\frac{-2\sqrt{7}\sqrt{7}-6\sqrt{7}-2\sqrt{7}-6}{-\sqrt{7}\sqrt{7}-3\sqrt{7}+3\sqrt{7}+9}$ oe **or** $\frac{-14-6\sqrt{7}-2\sqrt{7}-6}{-7+9}$ **or** $\frac{-20-8\sqrt{7}}{2}$ **or** $\frac{-20-\sqrt{448}}{2}$			M1 numerator correctly expanded and may be simplified to at least 2 terms and denominator correctly expanded and may be simplified to one term, this mark implies previous M1 $\frac{2\sqrt{7}+2}{\sqrt{7}-3}\times\frac{\sqrt{7}+3}{\sqrt{7}+3}=-10$ $-4\sqrt{7}$ scores M1M0
	Working required	$-10-\sqrt{112}$		A1 dep on M2 SCB1 for $-10-\sqrt{112}$ gained with no method marks awarded
				SCB2 for $-10-\sqrt{112}$ gained if you would award 1st M1 but not second M1 (total 2 marks)
				Total 3 marks

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Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 17

Q	Working	Answer	Mark	Notes
	$\frac{1+\sqrt{5}}{3-\sqrt{5}}\times\frac{3+\sqrt{5}}{3+\sqrt{5}}$ oe or $\frac{1+\sqrt{5}}{3-\sqrt{5}}\times\frac{-3-\sqrt{5}}{-3-\sqrt{5}}$ oe		3	M1 for rationalising the denominator by multiplying numerator and denominator by $3+\sqrt{5}$ (or $-3-\sqrt{5}$)
	$\frac{3+\sqrt{5}+3\sqrt{5}+\sqrt{5}\sqrt{5}}{9+3\sqrt{5}-3\sqrt{5}-\sqrt{5}\sqrt{5}}$ oe or $\frac{3+\sqrt{5}+3\sqrt{5}+\sqrt{5}\sqrt{5}}{9-\sqrt{5}\sqrt{5}}$ oe or $\frac{3+\sqrt{5}+3\sqrt{5}+5}{9+3\sqrt{5}-3\sqrt{5}-5}$ oe or $\frac{8+\sqrt{5}+3\sqrt{5}}{9-5}$ oe or $\frac{3+4\sqrt{5}+5}{9-5}$ oe or $\frac{8+4\sqrt{5}}{4}$			M1 numerator correctly expanded and may be simplified to at least 2 terms and denominator correctly expanded and may be simplified to 1 term
	Working required	$2+\sqrt{5}$		A1 for $2+\sqrt{5}$ from correct working dep on M2
				Total 3 marks

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Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 17

Question	Working	Answer	Mark	Notes
	$20-8\sqrt{5}-5\sqrt{20}+2\sqrt{5}\sqrt{20}$ **or** $20-8\sqrt{5}-5\sqrt{20}+2\sqrt{100}$ **or** $20-8\sqrt{5}-5\sqrt{20}+20$ **or** $20-8\sqrt{5}-10\sqrt{5}+20$ **or** $20-4\sqrt{20}-5\sqrt{20}+20$ **or** $(4-2\sqrt{5})(5-2\sqrt{5})$ **or** $\sqrt{20}=\sqrt{4}\sqrt{5}$ **or** $\sqrt{20}=2\sqrt{5}$ **or** $5\sqrt{20}=5(\sqrt{4\times5})$ **or** $5\sqrt{20}=10\sqrt{5}$		3	M1 for expansion of brackets with 3 out of 4 terms correct or 4 terms correct ignoring signs **or** for simplifying $\sqrt{20}$ may be seen before or after expansion of brackets **or** for writing $2\sqrt{5}$ as $\sqrt{20}$
	eg $20+20-18\sqrt{5}=40-18\sqrt{5}$ **or** $20-4\sqrt{20}-5\sqrt{20}+20=$ $40-9\sqrt{20}$			M1 dep on M1 for correct expansion of brackets **with** 3 or 4 terms shown in their expansion **and** correct simplification to two terms
	Working required	$40-\sqrt{1620}$		A1 (dep on M2) accept $a=40$ $b=1620$ SCB1 for $40-\sqrt{1620}$ gained if you would award 1st M1 but not 2nd M1 (total 2nd marks)
				Total 3 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2022 · Question 17

17			3	
	eg $\frac{\sqrt{12}}{\sqrt{3}+2}\times\frac{\sqrt{3}-2}{\sqrt{3}-2}$			M1 rationalise denominator – award for seeing multiplication by $\frac{\sqrt{3}-2}{\sqrt{3}-2}$ or $\frac{-\sqrt{3}+2}{-\sqrt{3}+2}$
	eg $\frac{\left(\sqrt{36}-2\sqrt{12}\right)}{3-4}$ **or** $\frac{6-2\sqrt{12}}{-1}$ **or** $-6+2\sqrt{12}$ **or** $\frac{6-4\sqrt{3}}{-1}$ **or** $-6+4\sqrt{3}$			M1 dep M1 correctly simplifying numerator and denominator. (denominator could be 3 – 4 or –1)
		$-6+\sqrt{48}$		A1 dep M2 must be in correct form (including $\sqrt{48}$) allow $a=-6$ and $b=48$
				Total 3 marks

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Surds - Mark scheme

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2021 · Question 17*

17	$\left(\frac{8}{\sqrt{5}-1}\right)\times\frac{\sqrt{5}+1}{\sqrt{5}+1}$ **or** $\frac{8\left(\sqrt{5}+1\right)}{4}$ **or** $\frac{8\sqrt{5}+8}{4}$ oe		3	M1 for rationalising the denominator – award for seeing intention to multiply by $\frac{\sqrt{5}+1}{\sqrt{5}+1}$ **or** $\frac{-\sqrt{5}-1}{-\sqrt{5}-1}$
	Working required	$2\sqrt{5}+2$		A1 from correct working
		$\sqrt{20}+2$		B1ft for $k\sqrt{5}+c=\sqrt{5k^2}+c$ where $5k$ is a single integer Accept $a = 20$ and $b = 2$
				Total 3 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2021 · Question 18*

Q	Working	Answer	Mark	Notes
	e.g. $40+8\sqrt{x}-5\sqrt{x}-\sqrt{x}\sqrt{x}$ **or** $40+8\sqrt{x}-5\sqrt{x}-\left(\sqrt{x}\right)^2$ **or** $40+8\sqrt{x}-5\sqrt{x}-x$ **or** $40+3\sqrt{x}-x$		3	M1 for a correct expansion with at least 3 out of 4 terms correct oe or all 3 terms correct
		$x = 19$		A1 (dep on M1) for $x = 19$
		$y = 3$		B1 for $y = 3$
				Total 3 marks

Question 23 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2019 · Question 18*

18	$\frac{\sqrt{8}}{\sqrt{8}-2}\times\frac{\sqrt{8}+2}{\sqrt{8}+2}$ $\frac{\sqrt{8}\left(\sqrt{8}+2\right)}{8-4}=\frac{8+2\sqrt{8}}{4}=\frac{8+4\sqrt{2}}{4}$ $=2+\sqrt{2}$	Shown	3	M1 or $\frac{2\sqrt{2}}{2\sqrt{2}-2}$ or $\frac{\sqrt{2}}{\sqrt{2}-1}$ M1 or $\frac{\sqrt{2}}{\sqrt{2}-1}\times\frac{\sqrt{2}+1}{\sqrt{2}+1}$ A1 (dep on M2) Conclusion - need not state the value of n

Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 19*

Q	Working	Answer	Mark	Notes
	$\frac{6}{3-\sqrt{7}}\times\frac{3+\sqrt{7}}{3+\sqrt{7}}$ or $\frac{6}{3-\sqrt{7}}\times\frac{-3-\sqrt{7}}{-3-\sqrt{7}}$			M1
	$\frac{6(3+\sqrt{7})}{3^2-7}$ **or** $\frac{6(3+\sqrt{7})}{2}$ **or** $\frac{6(-3-\sqrt{7})}{-3^2+7}$ **or** $\frac{6(-3-\sqrt{7})}{-2}$			M1 (numerator may be expanded or denominator may be 4 terms which need to be all correct)
		$9+3\sqrt{7}$	3	A1 dep on M2 for $9+3\sqrt{7}$ **or** $3(3+\sqrt{7})$ from correct working
				Total 3 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 15

Q	Working	Answer	Mark	Notes
a)		9	1	B1 allow $9\sqrt{3}$
(b)	$\frac{21}{3-\sqrt{2}}\times\frac{3+\sqrt{2}}{3+\sqrt{2}}$ **or** $\frac{21}{3-\sqrt{2}}\times\frac{-3-\sqrt{2}}{-3-\sqrt{2}}$		3	M1 for explicitly multiplying the numerator and the denominator by $3+\sqrt{2}$ or $-3-\sqrt{2}$
	eg $\frac{21\left(3+\sqrt{2}\right)}{9-3\sqrt{2}+3\sqrt{2}-2}$ **or** $\frac{21\left(3+\sqrt{2}\right)}{3^2-2}$ **or** $\frac{21\left(3+\sqrt{2}\right)}{9-2}$ **or** $\frac{21\left(3+\sqrt{2}\right)}{7}$ **or** $\frac{63+21\sqrt{2}}{9-2}$ **or** $\frac{63+21\sqrt{2}}{7}$			M1 dep on M1 (denominator may be 4 terms which all need to be correct) $\frac{21}{3-\sqrt{2}}\times\frac{3+\sqrt{2}}{3+\sqrt{2}}=9+3\sqrt{2}$ scores M1M0
	Working required	$9+\sqrt{18}$		A1 dep on M2 SCB1 for $9+\sqrt{18}$ gained with no method marks awarded SCB2 for $9+\sqrt{18}$ gained if you would award 1st M1 but not 2nd M1 (total 2 marks)
				Total 4 marks

Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2019 · Question 16

16	(a)	Eg $\frac{a+\sqrt{4b}}{a-\sqrt{4b}}\times\frac{a+\sqrt{4b}}{a+\sqrt{4b}}$ **or** $\frac{a+2\sqrt{b}}{a-2\sqrt{b}}\times\frac{a+2\sqrt{b}}{a+2\sqrt{b}}$ **or** $\frac{\left(a+2\sqrt{b}\right)^2}{\left(a+2\sqrt{b}\right)\left(a-2\sqrt{b}\right)}$			M1 For multiplying the numerator and denominator by $a+\sqrt{4b}$ or $a+2\sqrt{b}$
		Eg $\frac{(a+\sqrt{4b})(a+\sqrt{4b})}{a^2-4b}$			M1 dep on M1 for correctly simplified denominator
			$\frac{a^2+4a\sqrt{b}+4b}{a^2-4b}$	3	A1 for $\frac{a^2+4a\sqrt{b}+4b}{a^2-4b}$ **or** $\frac{\left(a+2\sqrt{b}\right)^2}{a^2-4b}$
	(b)		2.5 oe	1	B1

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Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 17

Q	Working	Answer	Mark	Notes
(a)		15	1	B1 accept k^{15}
(b)	eg $\frac{7(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})}$ or $\frac{7(-2-\sqrt{3})}{(2-\sqrt{3})(-2-\sqrt{3})}$		3	M1 for multiplying the numerator and denominator of the fraction by $2+\sqrt{3}$ or $-2-\sqrt{3}$
	eg $\frac{14+7\sqrt{3}}{4+2\sqrt{3}-2\sqrt{3}-3}$ or $\frac{14+7\sqrt{3}}{4-3}$ or $\frac{14+7\sqrt{3}}{1}$ or $\frac{-14-7\sqrt{3}}{-4-2\sqrt{3}+2\sqrt{3}+3}$ or $\frac{-14-7\sqrt{3}}{-4+3}$ or $\frac{-14-7\sqrt{3}}{-1}$			M1 dep on previous M1
	Working required	$\sqrt{147}+14$		A1 dep on M2 SCB1 for $\sqrt{147}+14$ gained with no method marks awarded SCB2 for $\sqrt{147}+14$ gained with 1st M1 awarded
				Total 4 marks

Question 28

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 17

Q	Working	Answer	Mark	Notes
(a)		$5\sqrt{27}$	1	B1 Allow $n = 5$ Do not accept 5 by itself
(b)	$\frac{5-\sqrt{2}}{\sqrt{2}-1}\times\frac{\sqrt{2}+1}{\sqrt{2}+1}$ or $\frac{5-\sqrt{2}}{\sqrt{2}-1}\times\frac{-\sqrt{2}-1}{-\sqrt{2}-1}$		3	M1 for rationalising the denominator by multiplying numerator and denominator by $\sqrt{2}+1$ or $-\sqrt{2}-1$
	eg $\frac{5\sqrt{2}+5-2-\sqrt{2}}{2-1}$ oe or $\frac{5\sqrt{2}+5-2-\sqrt{2}}{\sqrt{4}+\sqrt{2}-\sqrt{2}-1}$ oe or $5\sqrt{2}+5-2-\sqrt{2}$			M1 (numerator must be expanded to 4 terms, denominator may be 4 terms which need to be all correct) Accept 1 in the denominator without working
	Working required	$3+4\sqrt{2}$		A1 or for stating $a = 3$ and $b = 4$ dep on M2
				Total 4 marks

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Question 29

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 18

Q	Working	Answer	Mark	Notes
	$\sqrt{3}x - x = 6 + 2\sqrt{3}$ oe or $x - x\sqrt{3} = -6 - 2\sqrt{3}$ (allow $-2\sqrt{9}$ or $-2(\sqrt{3})^2$ for -6 or $2\sqrt{9}$ or $2(\sqrt{3})^2$ for 6)		4	M1 expanding bracket and collecting terms. Condone one error
	$(x =)\ \dfrac{6+2\sqrt{3}}{\sqrt{3}-1}$ oe eg $\dfrac{-6-2\sqrt{3}}{1-\sqrt{3}}$			A1 oe must be a correct fraction with irrational numerator and denominator
	$(x =)\ \dfrac{(6+2\sqrt{3})}{(\sqrt{3}-1)} \times \dfrac{(\sqrt{3}+1)}{(\sqrt{3}+1)}$ or $\dfrac{(6+2\sqrt{3})(\sqrt{3}+1)}{2}$ oe or $\dfrac{(6+2\sqrt{3})}{(-1+\sqrt{3})} \times \dfrac{(-1-\sqrt{3})}{(-1-\sqrt{3})}$ oe or $\dfrac{(-6-2\sqrt{3})(1+\sqrt{3})}{(1-\sqrt{3})(1+\sqrt{3})}$ oe			M1 (indep) Multiplying the numerator and denominator of their fraction by $\sqrt{3}+1$ oe or showing 2 or -2 as the denominator and multiplying the numerator by $\sqrt{3}+1$ oe or rationalising **their** denominator, so long as it is of the form $p+q\sqrt{3}$ where p and q are non zero integers (condone missing brackets provided meaning is clear)
	Working required	$6+4\sqrt{3}$		A1 dep on M1A1M1 with no errors seen
				Total 4 marks

Question 30

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 21

21	$(\sqrt{2}-1)^2 = 2-\sqrt{2}-\sqrt{2}+1\left(=3-2\sqrt{2}\right)$	$\dfrac{(3+\sqrt{8})}{(\sqrt{2}-1)^2} \times \dfrac{(\sqrt{2}+1)^2}{(\sqrt{2}+1)^2}$		4	M1 expand the denominator (accept $2-2\sqrt{2}+1$ - must see expansion) **OR** method to rationalise using $(\sqrt{2}+1)^2$
	$\dfrac{(3+\sqrt{8})}{"(3-2\sqrt{2})"} \times \dfrac{(3+2\sqrt{2})}{(3+2\sqrt{2})}$	$(\sqrt{2}-1)^2 = 2-\sqrt{2}-\sqrt{2}+1\left(=3-2\sqrt{2}\right)$ **or** $(\sqrt{2}+1)^2 = 2+\sqrt{2}+\sqrt{2}+1\left(=3+2\sqrt{2}\right)$ **or** $(\sqrt{2}-1)(\sqrt{2}+1) = 2-\sqrt{2}+\sqrt{2}-1\left(=1\right)$			M1 oe ft $3-2\sqrt{2}$ method to rationalise **OR** expansion of $(\sqrt{2}-1)^2$ (accept $2-2\sqrt{2}+1$) **or** $(\sqrt{2}+1)^2$ (accept $2+2\sqrt{2}+1$) **or** $(\sqrt{2}-1)(\sqrt{2}+1)$
	eg $\dfrac{9+6\sqrt{2}+3\sqrt{8}+8}{9-6\sqrt{2}+6\sqrt{2}-8}$ **or** $\dfrac{9+12\sqrt{2}+8}{9-8}$ **or** $\dfrac{9+6\sqrt{2}+3\sqrt{8}+8}{1}$ **or** $\dfrac{9+12\sqrt{2}+8}{1}$				M1 dep on 2nd M1 correct expansion of brackets
			$17+\sqrt{288}$		A1 or $p=17$, $q=288$ answer from fully correct working with intermediate steps of working seen
					Total 4 marks

CHALLENGE QUESTIONS

Question 31 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 23*

23	$(x^2 =)\ \dfrac{13+6\sqrt{5}}{2\sqrt{5}-3}$		4	M1	expression for x^2
	$\dfrac{13+6\sqrt{5}}{2\sqrt{5}-3}\times\dfrac{2\sqrt{5}+3}{2\sqrt{5}+3}$ or $\dfrac{13+6\sqrt{5}}{2\sqrt{5}-3}\times\dfrac{-2\sqrt{5}-3}{-2\sqrt{5}-3}$			M1	dep on previous M1 showing a **correct** product to rationalise the denominator (must be correct x^2)
	eg $\dfrac{13+6\sqrt{5}}{2\sqrt{5}-3}\times\dfrac{2\sqrt{5}+3}{2\sqrt{5}+3}=\dfrac{99+44\sqrt{5}}{11}$ or eg $\dfrac{13+6\sqrt{5}}{2\sqrt{5}-3}\times\dfrac{2\sqrt{5}+3}{2\sqrt{5}+3}=\dfrac{26\sqrt{5}+39+60+18\sqrt{5}}{20-9}$ or eg $\dfrac{13+6\sqrt{5}}{2\sqrt{5}-3}\times\dfrac{2\sqrt{5}+3}{2\sqrt{5}+3}=\dfrac{26\sqrt{5}+39+12(\sqrt{5})^2+18\sqrt{5}}{(2\sqrt{5})^2-3^2}$			M1	dep on previous M1 continuing the expansion of the product on the numerator and denominator – maybe one of these forms or a combination of forms
	Working required	$2+\sqrt{5}$		A1	dep on M3 accept $a = 2$, $b = 5$
					Total 4 marks

Question 32 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 20*

20	$\dfrac{18}{\sqrt{7}+1}\times\dfrac{\sqrt{7}-1}{\sqrt{7}-1}$		3	M1	for $\dfrac{18}{\sqrt{7}+1}\times\dfrac{\sqrt{7}-1}{\sqrt{7}-1}$
	eg $\dfrac{18(\sqrt{7}-1)}{7-1}$			M1	Dep on M1 for a correct numerator **and** multiplying out the denominator to 7 – 1 or 6
	$3\sqrt{7}-3$	$3\sqrt{7}-3$		A1	Dep on M2 Allow $3\ \sqrt{7}-1$
					Total 3 marks

Question 33 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 23*

23	**For 2 of** $30=5\times6$ or $30=5\times2\times3$ oe (for numerator) or $\sqrt{180}=6\sqrt{5}$ oe or $\sqrt{180}=2\times3\times\sqrt{5}$ (for denominator) or $25^{2x+7}=\left(5^2\right)^{2x+7}$ or $5^{2(2x+7)}$ oe or $\left(\sqrt{5}\right)^{4x+9}=\left(5^{\frac{1}{2}}\right)^{4x+9}$ or $5^{\frac{1}{2}(4x+9)}$ oe or $\dfrac{30}{\sqrt{180}}=\sqrt{5}$ oe or 5^{4x+15} as the numerator or 5^{2x+5} as the denominator		3	M1	or 5^{4x+15} as the numerator or 5^{2x+5} as the denominator or $30=5\times6$ or $30=5\times2\times3$ oe or $\sqrt{180}=6\sqrt{5}$ oe or $\sqrt{180}=2\times3\times\sqrt{5}$ or $25^{2x+7}=\left(5^2\right)^{2x+7}$ or $5^{2(2x+7)}$ oe or $\left(\sqrt{5}\right)^{4x+9}=\left(5^{\frac{1}{2}}\right)^{4x+9}$ or $5^{\frac{1}{2}(4x+9)}$ oe or
	$\dfrac{6\times5\times5^{4x+14}}{6\times5^{0.5}\times5^{2x+4.5}}$ oe eg $\dfrac{5^{4x+15}}{5^{2x+5}}$ or $\dfrac{\sqrt{5}^{8x+30}}{\sqrt{5}^{4x+10}}$ oe or $\dfrac{25^{2x+7.5}}{25^{x+2.5}}$ oe			M1	Correct expression in terms of 6 (or 2 and 3) and 5 with indices Some cancellation could have taken place **or** fully correct with $\sqrt{5}$ and powers **or** 25 and powers
	working required	5^{2x+10}		A1	dep on M2 allow $w = 2x + 10$
					Total 3 marks

Question 34

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 17

Question	Working	Answer	Mark	Notes
17 (a)		12	1	B1
(b)	$\frac{5-\sqrt{18}}{1-\sqrt{2}}\times\frac{1+\sqrt{2}}{1+\sqrt{2}}$ **or** $\frac{5-\sqrt{18}}{1-\sqrt{2}}\times\frac{-1-\sqrt{2}}{-1-\sqrt{2}}$ oe		3	M1 Multiplying numerator and denominator by $1+\sqrt{2}$
	$\frac{5-\sqrt{36}+5\sqrt{2}-\sqrt{18}}{1+\sqrt{2}-\sqrt{2}-2}$ **or** $\frac{5-6-3\sqrt{2}+5\sqrt{2}}{-1}$ **or** $\frac{-5+6+3\sqrt{2}-5\sqrt{2}}{1}$ oe NB:allow $\sqrt{18}$ or $3\sqrt{2}$ $\sqrt{36}$ or 6 or $\sqrt{6}\sqrt{6}$			M1 Showing correct expansions (not necessarily as a fraction)
	working required	$1-2\sqrt{2}$		A1 dep on M2 (ie all stages of working must be shown convincingly) or for stating $a = 1$ and $b = -2$
				Total 4 marks

Question 35

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 17

Question	Working	Answer	Mark	Notes
(a)	$6\times6+6\times2\sqrt{12}+6\times2\sqrt{12}+\left(2\times\sqrt{12}\right)^2$ or $36+12\sqrt{12}+12\sqrt{12}+4\sqrt{12}\sqrt{12}$ or $36+12\sqrt{12}+12\sqrt{12}+(4\times12)$ or $36+24\sqrt{3}+24\sqrt{3}+48$ or $36+2\times24\sqrt{3}+48$ or $36+6\times2\times2\sqrt{12}+48$		3	M1 for correct expansion of brackets showing **four** terms (need not be simplified) **or** for the use of $(a+b)^2 = a^2 + 2ab + b^2$ **or** for showing or stating $\sqrt{12} = 2\sqrt{3}$ oe
	$84+48\sqrt{3}$			M1 (dep on M1)
		Shown		A1 for fully correct working leading to given expression

Question	Working	Answer	Mark	Notes
(b)	E.g. $\left(\frac{3a^4}{t^5}\right)^{-2}$ or $\left(\frac{t^{15}}{27a^{12}}\right)^{\frac{2}{3}}$ or $\left(\frac{729a^{24}}{t^{30}}\right)^{-\frac{1}{3}}$		3	M1 for **one** of cube rooting **or** inverting **or** squaring or $\frac{ka^{-8}}{t^{-10}}$ where k is an integer $\neq 0$
	E.g. $\left(\frac{9a^8}{t^{10}}\right)^{-1}$ or $\frac{3^{-2}a^{-8}}{t^{-10}}$ or $\frac{\frac{1}{9}a^{-8}}{t^{-10}}$ or $\left(\frac{t^5}{3a^4}\right)^2$ or $\left(\frac{t^{30}}{729a^{24}}\right)^{\frac{1}{3}}$ or $\frac{a^{-8}}{9t^{-10}}$			M1 for **two** of cube rooting **or** inverting **or** squaring or $\frac{t^{10}}{ka^8}$ where k is an integer $\neq 0$
		$\frac{t^{10}}{9a^8}$		A1 Allow $\frac{t^{10}a^{-8}}{9}$ or $\frac{1}{9}t^{10}a^{-8}$
				Total 6 marks